

Create Capture History Objects

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2026-08-24

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Building Capture Histories for SCR Analyses

In this markdown, we'll create capture histories for the SCR analyses that follow. Each object will include i) a `y` array of capture histories (with `dim = c(i, j, k, t)`, individuals, traps, occasions, years), ii) a `pc` array (of matching dimensions) with previous capture information (to allow us to model a local behavioral effect of capture), and iii) a `clone` vector that has identifiers for each individual i (from the `finalClone` column of our `sampleMetadata` file). We build several of these objects, which would allow fitting different (open vs. closed) SCR models.

1. For closed population models - data for the two years are saved separately (in two 3-d arrays), also separate for the two sexes
 - Females in 2023
 - Females in 2024
 - Males in 2023
 - Males in 2024
2. For open population models - data for the two years are combined (4-d array), but we still keep sexes separate
 - Females (2023 and 2024)
 - Males (2023 and 2024)

Load data, source function

Load *i*) the sample metadata with clone ID information and *ii*) the `.Rda` with locations and yearly operational status of hair snares. Source in functions for *i*) building capture history and previous capture objects and *ii*) calculating summaries of capture histories from a 3-d or 4-d `y` array (based on a function from the [scrbook R package](#); Royle et al. 2017).

```

#Load sample metadata (smd)
smd <- read.csv("../sampleinfo/sampleMetaData_cloneID.csv", header = TRUE)
#Load snare location information
load("../sampleinfo/snareLocations.Rda")
ls()

```

```
## [1] "locs" "oper" "smd"
```

```

#Source the makeCH function
source("../input/makeCH.R")
#Source the sumstats functions (for 3d and 4d arrays)
source("../input/sumstats_4d.R")

```

Here's the function for making a capture history...

```
writeLines(readLines("../input/makeCH.R"))
```

```
## Warning in readLines("../input/makeCH.R"): incomplete final line found on
## '../input/makeCH.R'
```

```

## makeCH <- function(samps=NULL, traps=NULL, yrs=NULL) {
##   #Make an ID for the capture history (row index)
##   clone <- unique(samps$finalClone) #The finalClone identifier!
##
##   chID <- c()
##   for(i in 1:nrow(samps)) {
##     chID[i] <- which(clone == samps$finalClone[i])
##   }
##
##   if(length(yrs) == 1) {
##     pc <- y <- array(0, dim=c(max(chID),
##                               nrow(traps),
##                               max(samps$week)))
##     for(i in 1:nrow(samps)) {
##       y[chID[i],
##         which(rownames(traps) == samps$utmSnare[i]),
##         as.numeric(samps$week[i])] <- 1
##     }
##     #Now make the previous capture object
##     for(i in 1:dim(y)[1]){
##       for(j in 1:dim(y)[2]) {
##         if(sum(y[i,j,]) > 0) {
##           ij.capt1 <- which(y[i,j,] == 1)[1]
##           if(ij.capt1 < dim(y)[3]) {
##             pc[i,j,(ij.capt1+1):dim(y)[3]] <- 1
##           }
##         }
##       }
##     }
##   } else {
##     pc <- y <- array(0, dim=c(max(chID),
##                               nrow(traps),

```

```

##                                     max(samps$week),
##                                     length(yrs)))
##
##   for(i in 1:nrow(samps)) {
##     y[chID[i],
##       which(rownames(traps) == samps$utmSnare[i]),
##       as.numeric(samps$week[i]),
##       which(yrs == samps$year[i])] <- 1
##   }
##
##   #Now make the previous capture object
##   occ_cntr <- data.frame(occno = c(1:(dim(y)[3]*dim(y)[4])),
##                         secocc = rep(c(1:dim(y)[3]),dim(y)[4]),
##                         priocc = sort(rep(1:dim(y)[4],dim(y)[3])))
##
##   for(i in 1:dim(y)[1]){
##     for(j in 1:dim(y)[2]) {
##       if(sum(y[i,j,,]) > 0) {
##         ij.capt1 <- which(y[i,j,,] == 1, arr.ind = TRUE)[1,]
##         ij.capt1occno <- occ_cntr$occno[ij.capt1[1] == occ_cntr$secocc &
##                                         ij.capt1[2] == occ_cntr$priocc]
##         if(ij.capt1occno < nrow(occ_cntr)) {
##           for(r in (ij.capt1occno+1):nrow(occ_cntr)) {
##             pc[i,
##               j,
##               occ_cntr[r,2],
##               occ_cntr[r,3]] <- 1
##           }
##         }
##         rm(ij.capt1, ij.capt1occno)
##       }
##     }
##   }
##
##   out <- list(y, pc, clone)
##   names(out) <- c("y","pc","clone")
##   return(out)
## }

```

Closed population models

Splitting the dataset up by year and sex...

```

#Start with all hair samples from 2023 and 2024
sampsALL <- smd[which(!is.na(smd$finalClone) &
                      smd$is.SCR == TRUE &
                      smd$year %in% c(2023,2024) &
                      smd$FILT == FALSE &
                      !is.na(smd$finalSEX)),]

#Check the dataset

```

```
nrow(sampsALL)
```

```
## [1] 1473
```

```
table(sampsALL$region)
```

```
##  
## CGA  
## 1473
```

```
table(sampsALL$finalSEX)
```

```
##  
## F M  
## 775 698
```

```
sum(!sampsALL$finalSEX %in% c("M", "F"))
```

```
## [1] 0
```

```
length(unique(sampsALL$finalClone[!is.na(sampsALL$finalClone)]))
```

```
## [1] 165
```

```
sum(is.na(sampsALL$finalClone))
```

```
## [1] 0
```

```
#Separately for males, females, 2023, and 2024
```

```
snare23 <- locs[oper[,1] == 1,]
```

```
snare24 <- locs[oper[,2] == 1,]
```

```
#M23
```

```
Msamps23 <- sampsALL[which(sampsALL$year == 2023 &  
                           sampsALL$finalSEX == "M"),]
```

```
dim(Msamps23)
```

```
## [1] 249 20
```

```
table(Msamps23$finalSEX)
```

```
##  
## M  
## 249
```

```
table(Msamps23$year)
```

```
##  
## 2023  
## 249
```

```
length(unique(Msamps23$finalClone[!is.na(Msamps23$finalClone)]))
```

```
## [1] 55
```

```
M23 <- makeCH(samps=Msamps23,  
              traps=snares23,  
              yrs=2023)  
dim(M23$y)
```

```
## [1] 55 130 7
```

```
dim(M23$pc)
```

```
## [1] 55 130 7
```

```
length(M23$clone)
```

```
## [1] 55
```

```
#M24  
Msamps24 <- sampsALL[which(sampsALL$year == 2024 &  
                           sampsALL$finalSEX == "M"),]  
dim(Msamps24)
```

```
## [1] 449 20
```

```
length(unique(Msamps24$cloneID[!is.na(Msamps24$cloneID)]))
```

```
## [1] 0
```

```
M24 <- makeCH(samps=Msamps24,  
              traps=snares24,  
              yrs=2024)  
dim(M24$y)
```

```
## [1] 37 138 7
```

```
dim(M24$pc)
```

```
## [1] 37 138 7
```

```
length(M24$clone)
```

```
## [1] 37
```

```
length(unique(c(M23$clone,M24$clone)))
```

```
## [1] 77
```

```
#F23  
Fsamps23 <- sampsALL[which(sampsALL$year == 2023 &  
                           sampsALL$finalSEX == "F"),]  
dim(Fsamps23)
```

```
## [1] 343 20
```

```
length(unique(Fsamps23$finalClone[!is.na(Fsamps23$finalClone)]))
```

```
## [1] 43
```

```
F23 <- makeCH(samps=Fsamps23,  
              traps=snares23,  
              yrs=2023)  
dim(F23$y)
```

```
## [1] 43 130 7
```

```
dim(F23$pc)
```

```
## [1] 43 130 7
```

```
length(F23$clone)
```

```
## [1] 43
```

```
#F24  
Fsamps24 <- sampsALL[which(sampsALL$year == 2024 &  
                           sampsALL$finalSEX == "F"),]  
dim(Fsamps24)
```

```
## [1] 432 20
```

```
length(unique(Fsamps24$finalClone[!is.na(Fsamps24$finalClone)]))
```

```
## [1] 70
```

```
F24 <- makeCH(samps=Fsamps24,
              traps=snares24,
              yrs=2024)
dim(F24$y)
```

```
## [1] 70 138 7
```

```
dim(F24$pc)
```

```
## [1] 70 138 7
```

```
length(F24$clone)
```

```
## [1] 70
```

```
length(unique(c(F23$clone, F24$clone)))
```

```
## [1] 88
```

```
#And the summary stats
rowSums(M23$y)
```

```
## [1] 3 5 7 8 1 1 3 1 7 1 3 2 1 5 1 1 2 4 1 4 1 1 1 2 2 1 7 2 2 1 1 1 1 3 1 1 1 1
## [39] 1 1 1 1 1 2 1 1 1 4 2 1 2 1 2 2 1
```

```
sumstats_3d(M23$y)
```

```
##          ss.totN      ss.totcaps      ss.avgcaps ss.indwsprecap      ss.avgtraps
##          55.000000      116.000000          2.109091          21.000000          1.690909
```

```
rowSums(M24$y)
```

```
## [1] 1 26 25 5 5 9 4 6 5 4 1 13 1 1 3 1 5 1 1 1 1 1 4 1 1
## [26] 1 1 1 1 1 1 1 1 1 1 1 1 1
```

```
sumstats_3d(M24$y)
```

```
##          ss.totN      ss.totcaps      ss.avgcaps ss.indwsprecap      ss.avgtraps
##          37.000000      138.000000          3.729730          13.000000          2.540541
```

```
rowSums(F23$y)
```

```
## [1] 6 2 5 15 9 3 6 6 5 3 6 3 2 1 2 1 3 1 1 1 1 1 1 1 1 1
## [26] 3 1 2 1 1 1 2 1 2 1 1 1 1 1 1 1 1
```

```
sumstats_3d(F23$y)
```

```
##          ss.totN      ss.totcaps    ss.avgcaps ss.indwsprecap    ss.avgtraps
##      43.000000      109.000000      2.534884      13.000000      1.511628
```

```
rowSums(F24$y)
```

```
## [1]  9  5  6  7 10  2  6  2  1  5  3  1  2  5  2  2  1  2  4  2  1  1  1  1  6
## [26]  1  1  3  1  1  1  1  1  1  2  1  2  1  4  1  1  3  1  1  2  1  1  1  1  1
## [51]  2  1  2  1  2  1  1  2  1  4  1  2  1  1  2  1  1  1  1  1  1
```

```
sumstats_3d(F24$y)
```

```
##          ss.totN      ss.totcaps    ss.avgcaps ss.indwsprecap    ss.avgtraps
##      70.000000      151.000000      2.157143      19.000000      1.400000
```

Open population models

Combining data for years, but sexes are still separate...

```
#Females - both years
Fsamps <- sampsALL[which(sampsALL$finalSEX == "F"),]
length(unique(Fsamps$finalClone[!is.na(Fsamps$finalClone)]))
```

```
## [1] 88
```

```
Fch <- makeCH(samps=Fsamps,
              traps=locs,
              yrs=c(2023,2024))
dim(Fch$y)
```

```
## [1]  88 154   7   2
```

```
dim(Fch$pc)
```

```
## [1]  88 154   7   2
```

```
length(Fch$clone)
```

```
## [1] 88
```

```
#Males - both years
Msamps <- sampsALL[which(sampsALL$finalSEX == "M"),]
length(unique(Msamps$finalClone[!is.na(Msamps$finalClone)]))
```

```
## [1] 77
```

```
Mch <- makeCH(samps=Msamps,
              traps=locs,
              yrs=c(2023,2024))
```

```
dim(Mch$y)
```

```
## [1] 77 154 7 2
```

```
dim(Mch$pc)
```

```
## [1] 77 154 7 2
```

```
length(Mch$clone)
```

```
## [1] 77
```

```
#And the summary stats
rowSums(Fch$y)
```

```
## [1] 13 8 14 25 5 9 3 7 6 7 7 5 3 6 1 5 5 4 1 6 2 5 3 2 4
## [26] 3 3 2 2 1 1 1 1 6 2 1 5 2 5 1 1 1 1 1 1 2 1 1 2 1
## [51] 1 3 1 1 1 1 3 1 1 2 2 2 2 1 1 1 2 1 2 1 1 2 1 1 2
## [76] 1 5 1 2 1 1 2 1 1 1 1 1 1
```

```
sumstats_4d(Fch$y)
```

```
##      ss.totN      ss.totcaps      ss.avgcaps ss.indwsprecap      ss.avgtraps
##      88.000000      260.000000      2.954545      31.000000      1.602273
##      ss.avgyrs
##      1.284091
```

```
rowSums(Mch$y)
```

```
## [1] 1 8 14 13 33 1 1 3 1 12 1 29 2 1 6 1 1 2 7 7 1 4 1 5 4
## [26] 1 3 2 1 7 2 5 15 1 1 1 1 1 5 1 1 1 3 1 1 1 1 1 1 1
## [51] 1 1 4 1 1 1 1 1 1 1 1 2 1 1 1 2 1 1 2 3 1 1 1 1 1
## [76] 1 2
```

```
sumstats_4d(Mch$y)
```

```
##      ss.totN      ss.totcaps      ss.avgcaps ss.indwsprecap      ss.avgtraps
##      77.000000      254.000000      3.298701      26.000000      2.246753
##      ss.avgyrs
##      1.194805
```

Save the output

Save all of these objects (F23,F24,M23,M24,Fch,Mch) to a single .Rda file...

```
save(F23, F24, M23, M24, Fch, Mch, file = "CGA_CH.Rda")
```
